Repolar Diagnostics

AI risk scores for inherited arrhythmias and drug-induced QT prolongation

Legal form: SAS (Lyon, FR) **Founded:** 2022 **Employees:** 21

Stage: Seed+ (EUR 6 M, closed Q1 2026) **Contact:** Partnerships, Repolar Diagnostics

1. Executive summary

Repolar Diagnostics develops **RepoScore**, software that estimates the risk of life-threatening arrhythmias in patients with inherited channelopathies such as long QT syndrome (LQTS) and short QT syndrome (SQTS), from 12-lead and wearable ECG recordings combined with genetic findings. A second product, **RepoTox**, screens drug candidates for QT prolongation and cardiotoxicity for pharmaceutical companies.

We want to open a **Swiss research and validation site** in 2027 and are evaluating Canton Bern, because of its cardiovascular research and its university hospital.

What we are looking for in Bern

- Research groups in **cardiac electrophysiology and genetic arrhythmia syndromes** (LQTS, SQTS, arrhythmogenic cardiomyopathy).
- Experimental models of arrhythmia, such as **transgenic animal models**, to validate our risk markers.
- Expertise in **cardiotoxicity testing** with human iPSC-derived cardiomyocytes for RepoTox.
- Clinical partners for a registry study on **personalized risk stratification** of channelopathy patients.
- Know-how in pharmacology of **anti-arrhythmic drugs** and drug effects on cardiac repolarization.

2. Problem

- Inherited arrhythmia syndromes affect about 1 in 2,000 people and are a leading cause of sudden cardiac death in the young.
- Today's risk scores use the QT interval alone and miss many patients at risk, while others get defibrillators they never need.
- About one in ten drug candidates fails in development because of cardiac safety findings, often late and at high cost.

3. Solution and technology

RepoScore

- Deep-learning model on ECG morphology, heart-rate dynamics and genotype.
- Works with 12-lead ECGs and with single-lead wearable recordings.
- Retrospective validation on 3,400 patients: AUC 0.86 for arrhythmic events.

RepoTox

- In-vitro assay on human iPSC-derived cardiomyocytes with field-potential recordings.
- Combined with in-silico action-potential models of repolarization.
- Service for pharma: report within 3 weeks per compound.

4. Market and business model

Product	Customer	Price
RepoScore licence	Cardiology clinics, ECG device makers	CHF 18,000 / clinic / year
RepoTox screening	Pharmaceutical and biotech companies	CHF 25,000 / compound

Competitors: classic QT-based risk calculators and contract research organisations offering hERG assays only.

5. Plans for Switzerland

- **Team:** 12 people in Bern by 2028 (data scientists, electrophysiologists, lab technicians).
- **Space:** 250 m² offices and a BSL-1 cell culture lab.
- **Regulatory path:** RepoScore as software as a medical device (MDR class IIa), CE marking planned for 2028.

6. Team

- **CEO:** co-founder, former head of cardiac safety at a pharmaceutical company.
- **CSO:** co-founder, cardiologist and electrophysiologist, 40 publications on channelopathies.
- **Head of AI:** PhD in signal processing of ECG data.

Fictional company and figures, created as a demo input for the HackApertus Bern Fit prototype. Any resemblance to real companies is coincidental.